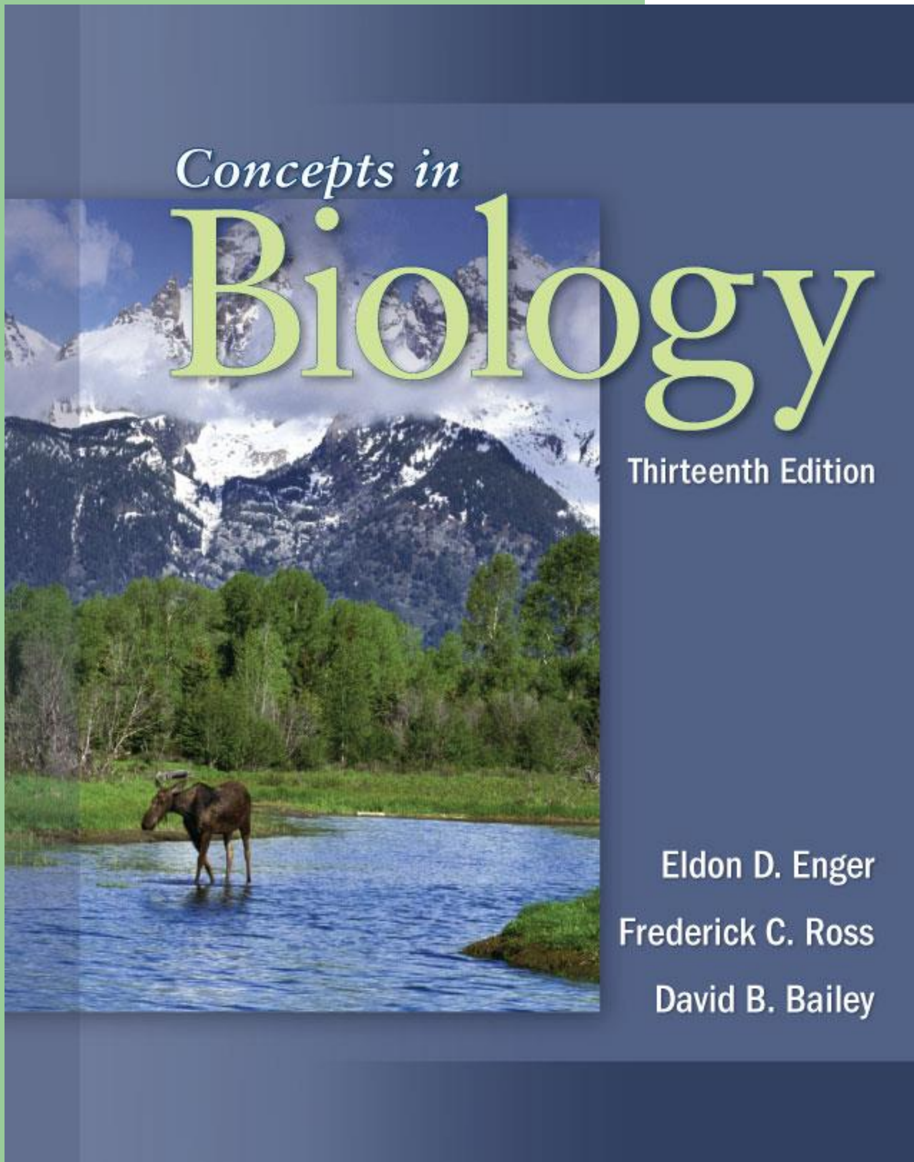
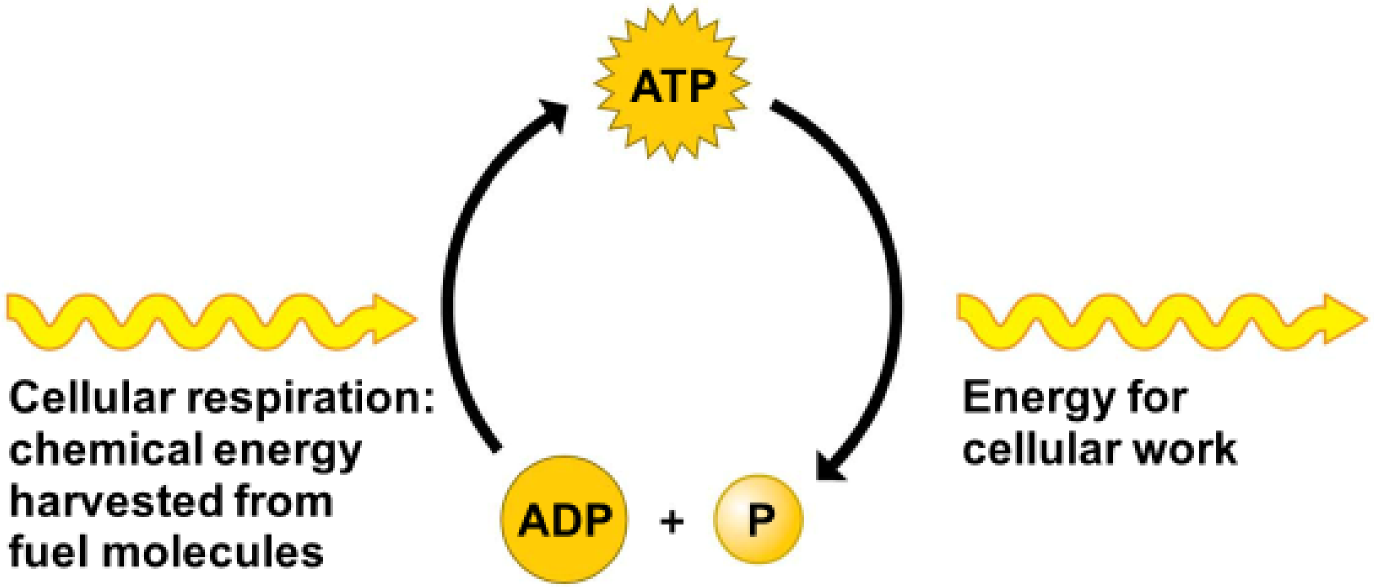


Lecture 5

Chapter 6:

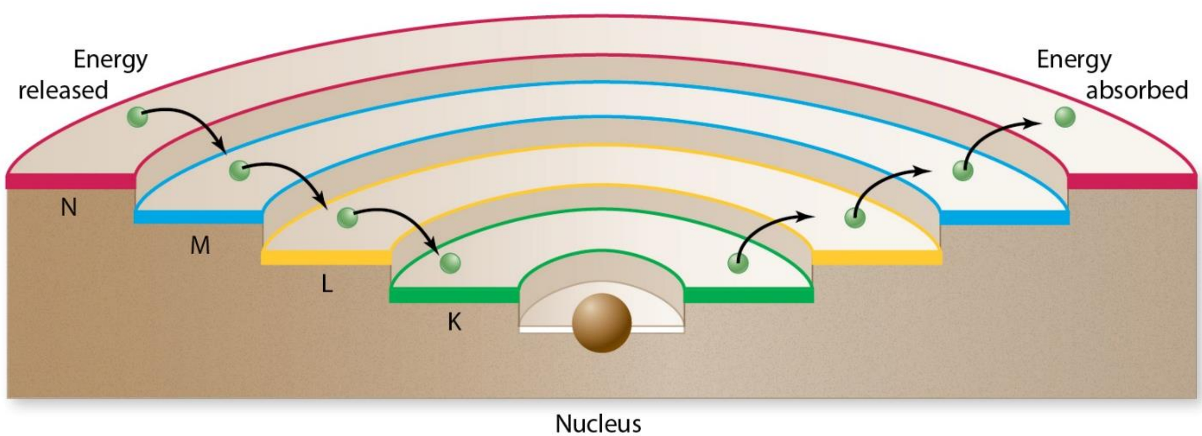
Cellular Respiration

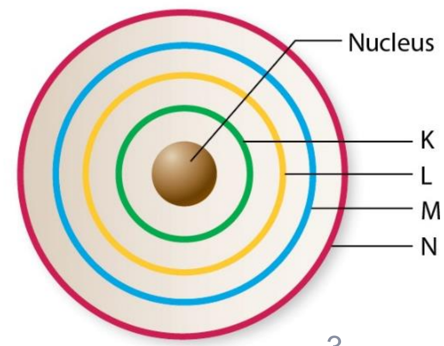
Reminder

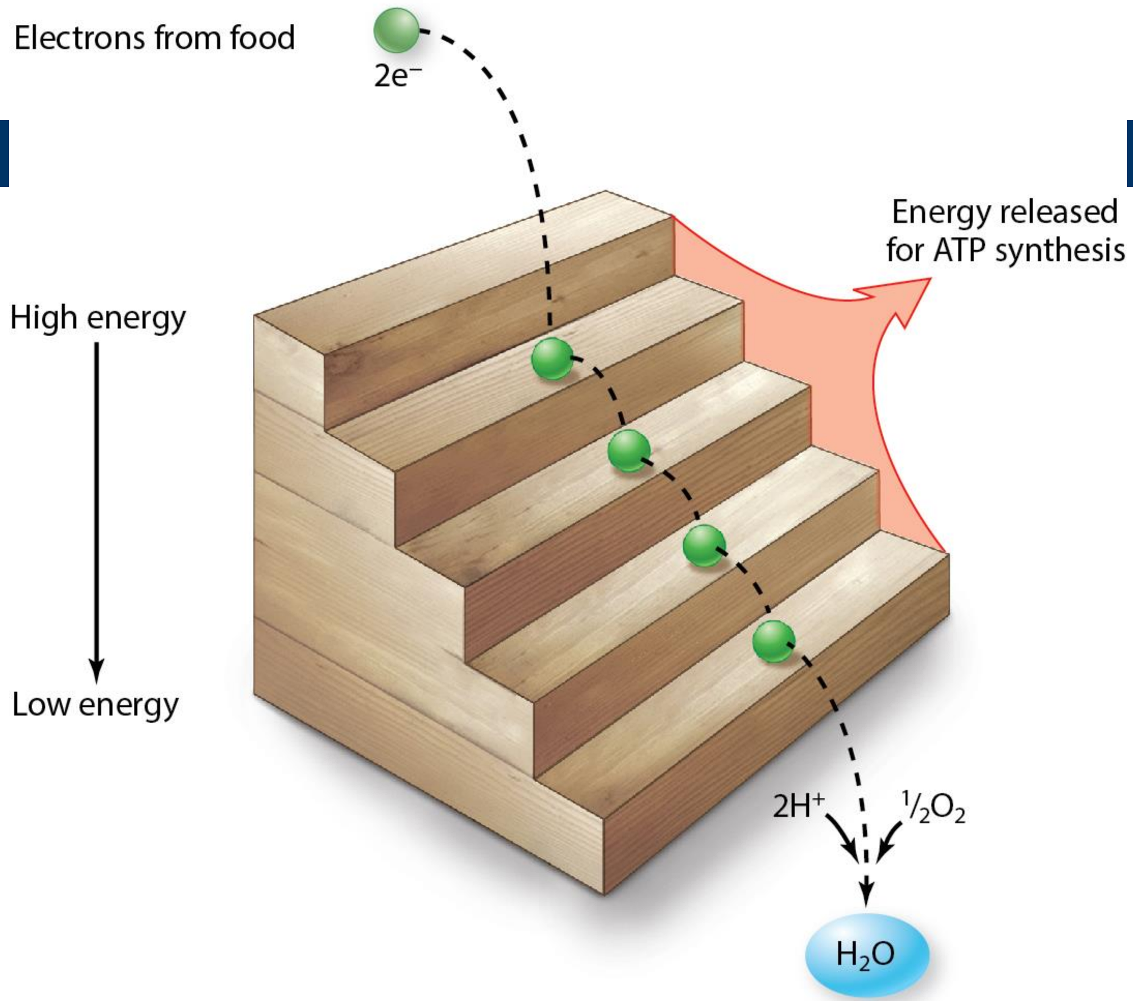

ATP
ADP
+
P
Cellular respiration: chemical energy harvested from fuel molecules
Energy for cellular work

Electrons and Energy

- Electrons have a potential energy that is related to their position ("energy of position")
- When an atom absorbs energy, an electron moves to a higher energy level, farther from the nucleus
- **When an electron falls to lower energy levels (by losing electrons), energy is released**






Electrons from food
2e⁻
Energy released for ATP synthesis
High energy
Low energy
2H+
1/2O2
H2O

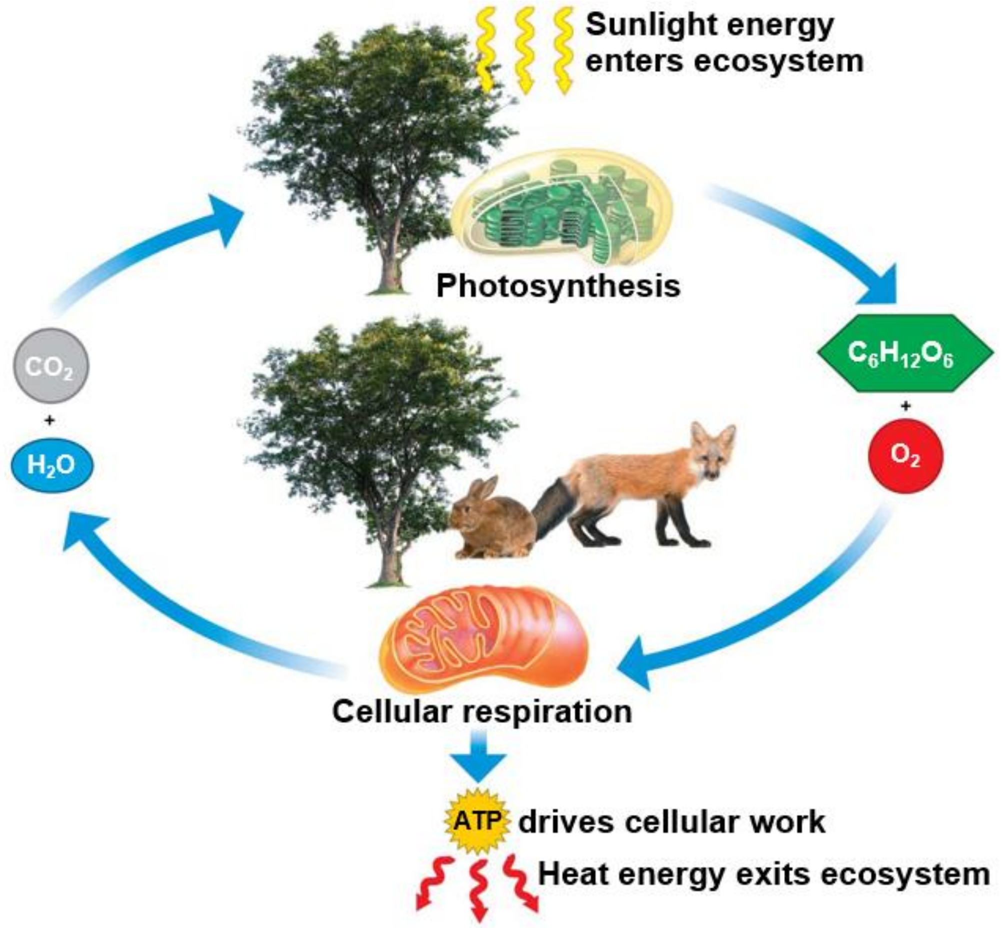
Oxidation-Reduction

- **Oxidation** = loss of electrons → electrons are picked up by **electron carriers** which become reduced
  - Electron carriers include NAD^+ and FAD
- **Reduction** = gain of electrons

Energy and Organisms

- Organisms are classified based on the kind of energy they use.
  - **Autotrophs**
    - Use the energy from sunlight to make organic molecules (sugar)
    - Use the energy in the organic molecules to make ATP
  - **Heterotrophs**
    - Obtain organic molecules by eating the autotrophs
    - Use the energy in the organic molecules to make ATP
- Autotrophs use **photosynthesis**.
  - To use the energy from light to make organic molecules
- All organisms use **cellular respiration**.
  - To harvest the energy from organic molecules and use it to make ATP

Energy Transformation



Cellular Respiration

- Some organisms require oxygen to perform cellular respiration ➔ **aerobic respiration**
- Other organisms carry out a form of respiration that does not require oxygen ➔ **anaerobic respiration**

Aerobic Respiration: An Overview

- **Aerobic cellular respiration** is a specific series of enzyme-controlled chemical reactions
  - Oxygen is involved in the **breakdown** of **glucose** into $\mathbf{CO_2}$ and $\mathbf{H_2O}$ ➔ chemical-bond energy released to the cell in the form of **ATP**

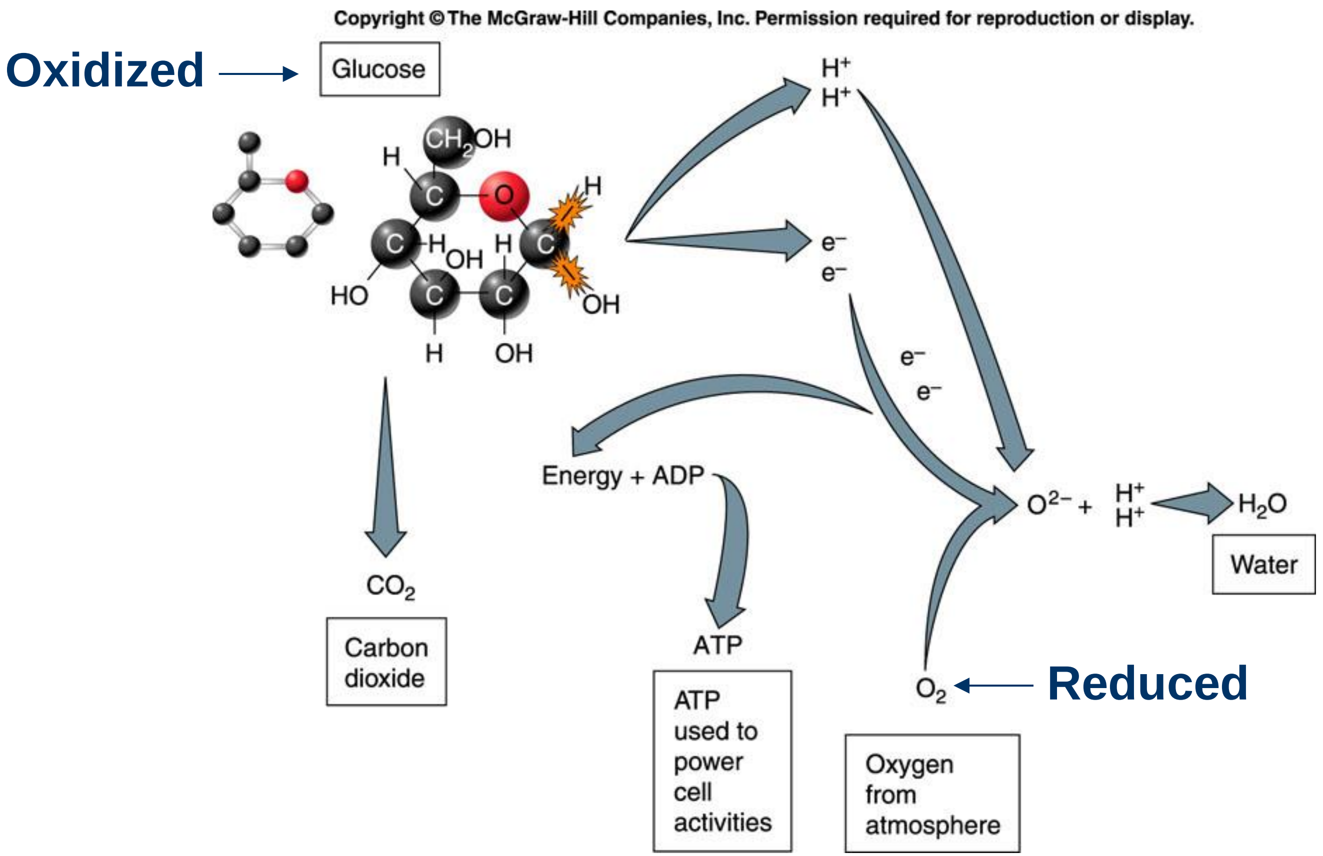
glucose	+ oxygen	→	carbon dioxide	+ water	+ energy
$C_6H_{12}O_6$	$+ 6\ O_2$	→	$6\ CO_2$	$+ 6\ H_2O$	+ energy (ATP + heat)

  - Covalent bonding = sharing of electrons ➔ when electrons are removed, glucose becomes **oxidized**➔ covalent bonds broken down

Aerobic Respiration: An Overview

- The high energy electrons released from the breakdown of glucose (oxidation of glucose) are transferred to electron carriers or **electron-transfer molecules**, NAD^+ and FAD.
  - Hold electrons temporarily before passing them on to other molecules ➔ **energy released** by the transfer of electrons ➔ **ATP** formed ➔ the electrons must be placed in a safe location ➔ they are added to **oxygen** ➔ becomes **reduced** (O^{2-})
  - So, in aerobic cellular respiration, glucose is oxidized and oxygen is reduced
  - The H^+ ions released from the oxidation of glucose combine with O^{2-} to form **H_2O**
  - The C and O atoms left from the breakdown of glucose combine to form **CO_2**

Aerobic Respiration and Oxidation-Reduction Reactions



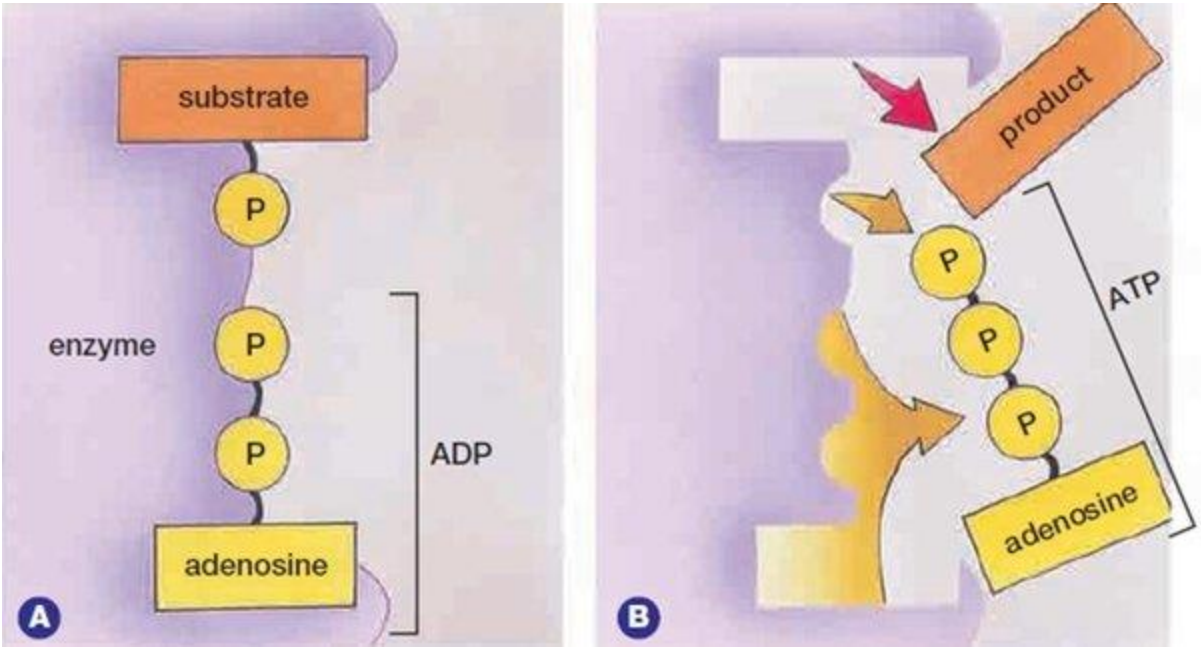
Aerobic Respiration: An Overview

- Of all the covalent bonds in glucose (O—H, C—H, C—C), those easiest to get at are the C—H and O—H bonds on the outside of the molecule. When these bonds are broken, two things happen:

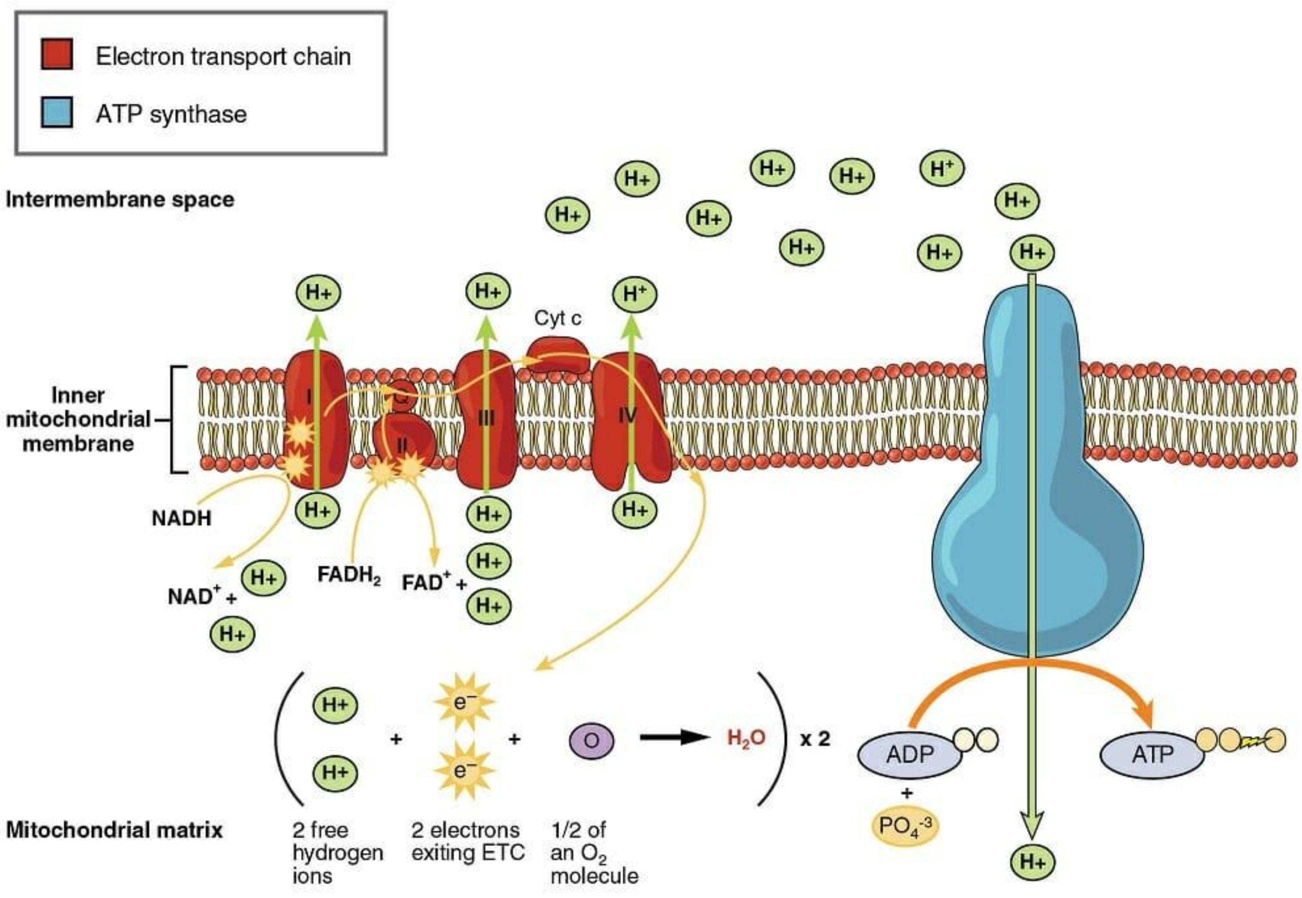
1. The energy of the electrons can ultimately be used to **phosphorylate ADP molecules to produce higher-energy ATP** molecules ➔ substrate-level phosphorylation.

2. **Hydrogen ions (protons)** are released and pumped across membranes, creating a **gradient**. When they flow back to the side from which they were pumped, their energy is used to generate even more ATP (most efficient) ➔ oxidative phosphorylation

Substrate-Level Phosphorylation



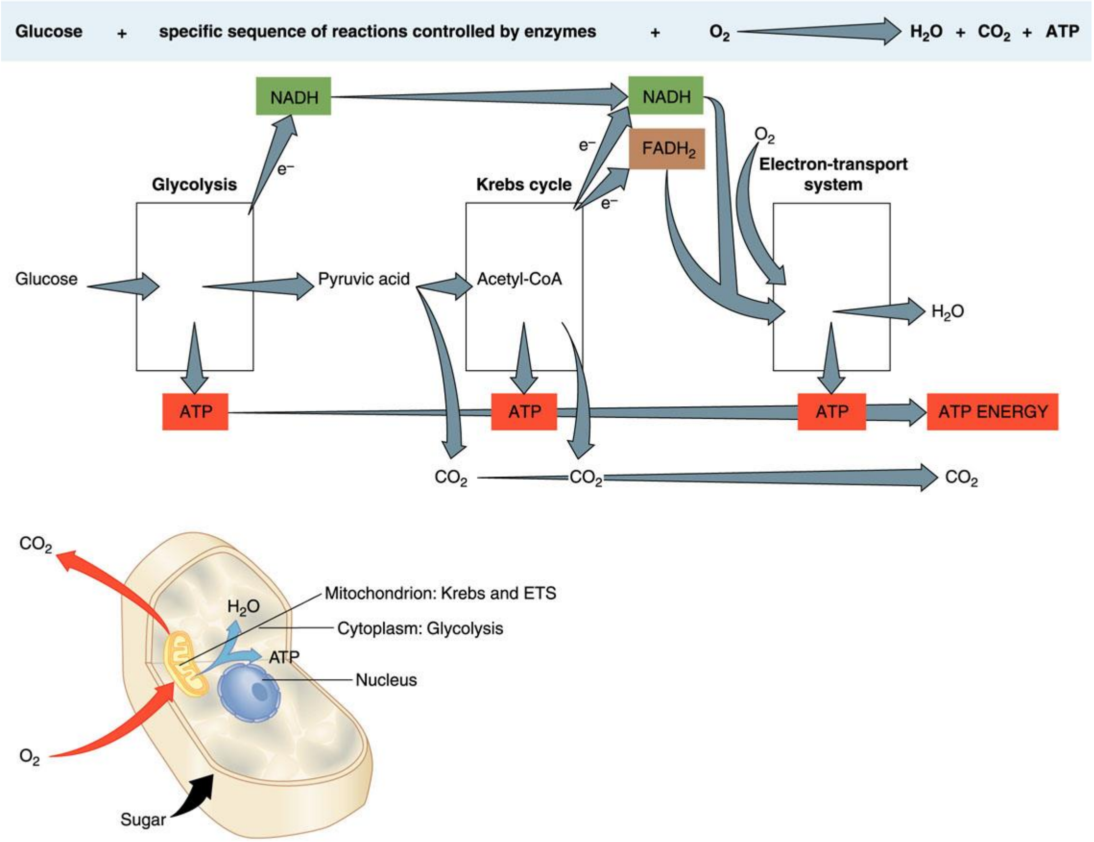
Oxidative Phosphorylation



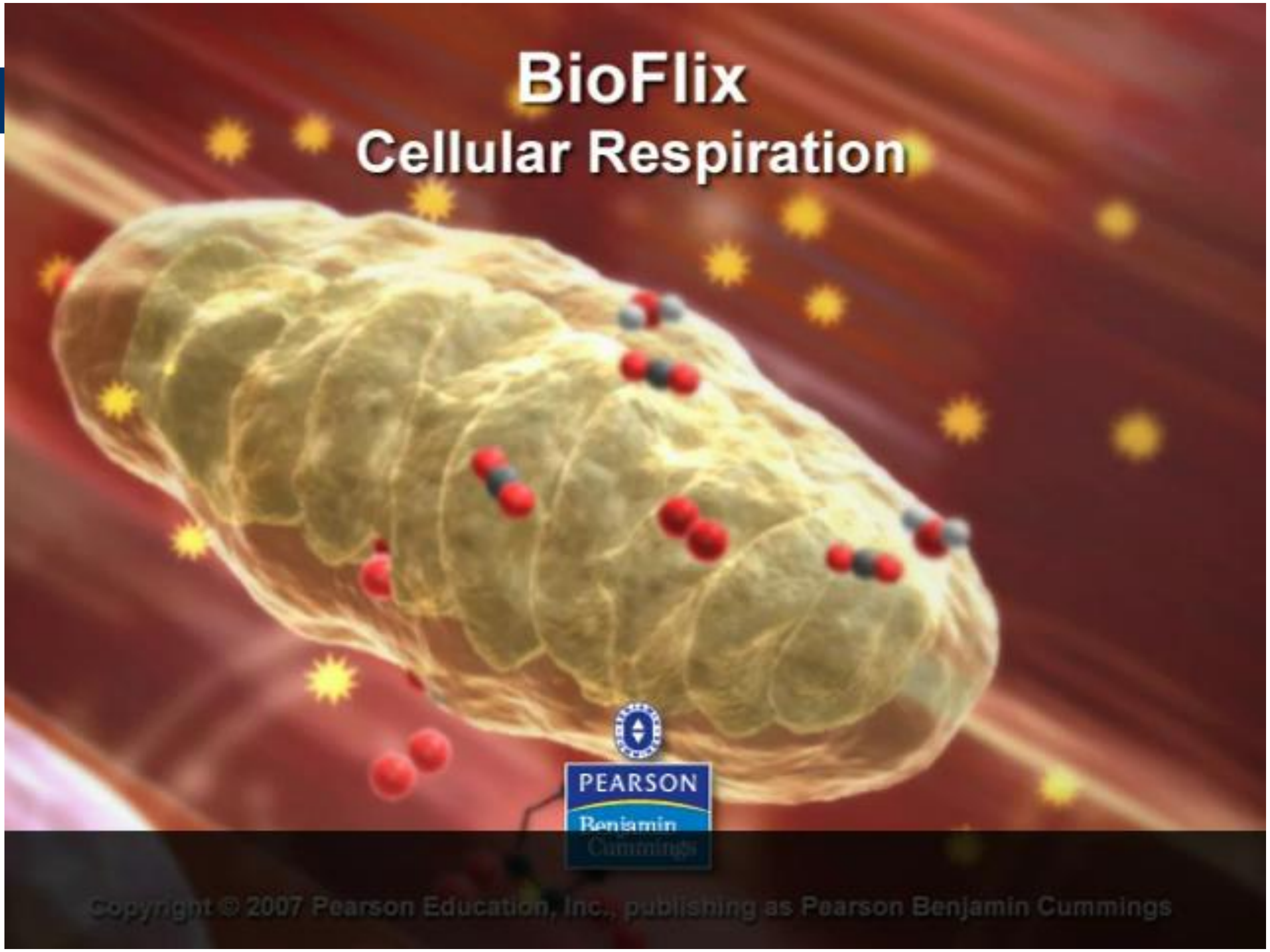
Aerobic Respiration: An Overview

- In eukaryotic cells, the process of releasing energy from food molecules **begins in the cytoplasm** and **is completed in the mitochondria**
- Three distinct enzymatic pathways involved:
  - **Glycolysis**
  - **Krebs cycle**
  - **Electron-transport system**

Aerobic Cellular Respiration: Overview



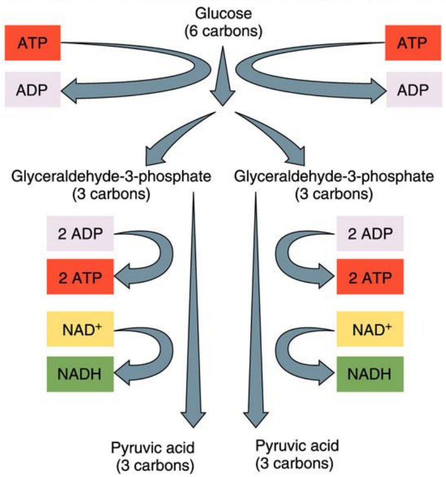
Aerobic Cellular Respiration: Overview



Glycolysis

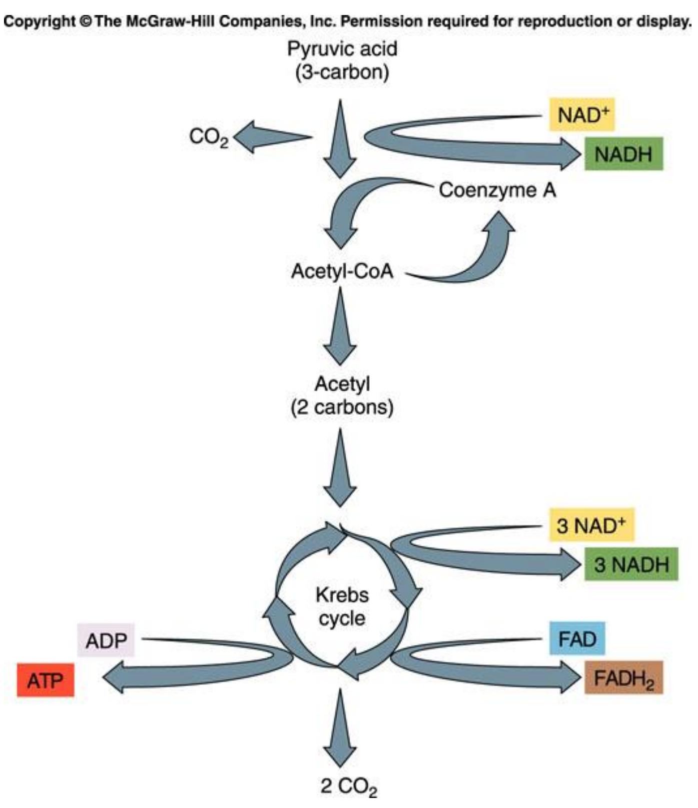
- A series of enzyme-controlled, **anaerobic** reactions that occur in the **cytoplasm**
- The **6-carbon glucose** is split into two smaller 3-carbon molecules known as **glyceraldehyde-3-phosphate** which undergo additional reactions to form **pyruvic acid**
- **Two ATP molecules are** used to split glucose
- As glucose is metabolized, enough energy is released to
  - Make 4 ATP molecules
    - 4 ATP made -2 ATP used =

      net production of 2 ATP
  - 2 NAD+ are reduced to NADH
    - The electrons picked up by NAD+ are transferred to the electron-transport system (ETS)
    - Once all electrons are dropped, NAD+ are available to pick more electrons and repeat the job



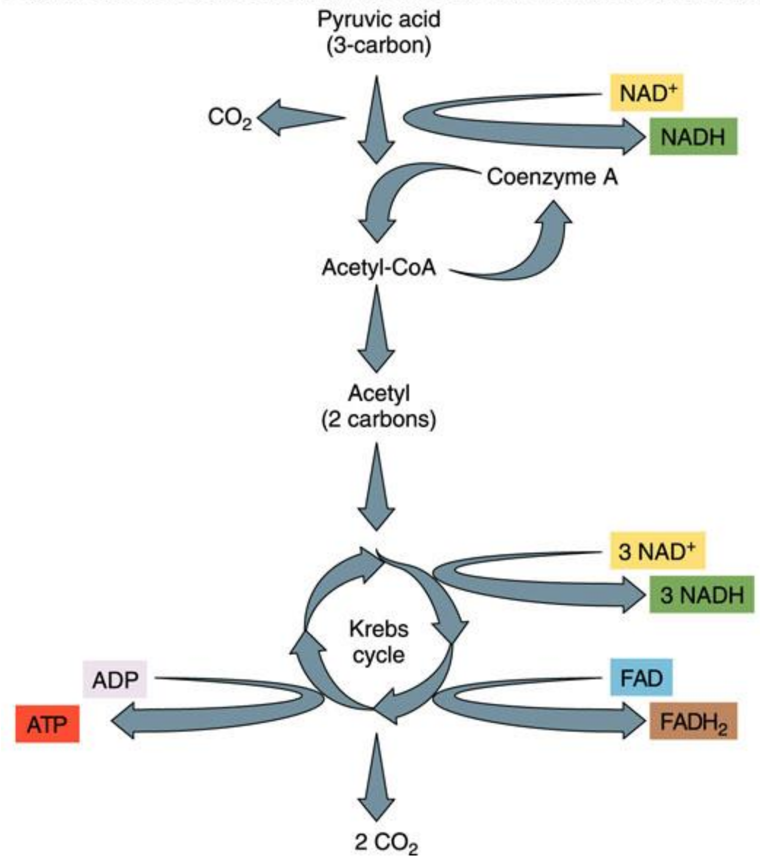
Acetyl CoA Formation

- Of the 3 carbons of pyruvic acid, one is tripped off and the remaining 2- carbon fragment is attached to a molecule of coenzyme A (CoA), becoming a compound called **acetyl-CoA ➔ enters the Kreb's cycle**



Kreb's Cycle

- Also known as the citric acid cycle or the tricarboxylic acid (TCA) cycle
- Takes place within the matrix of **mitochondria**



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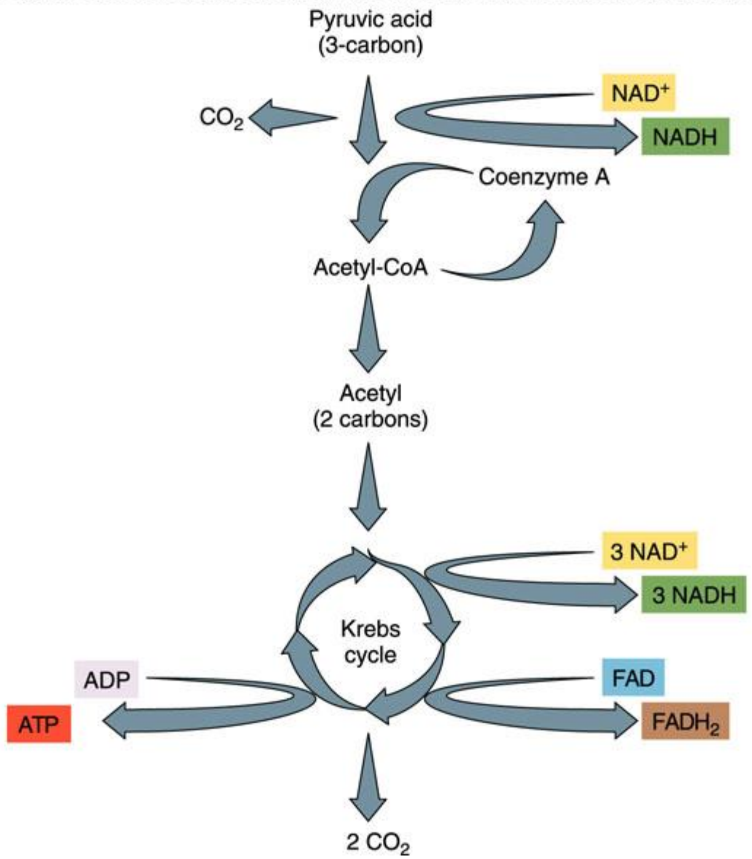
Kreb's Cycle

- The breakdown of pyruvic acid
  - Released as **carbon dioxide**
- Enough energy is released as one pyruvic acid molecule is metabolized to
  - **Make 1 ATP**
  - **Reduce 3 NAD+ to form 3 NADH**
  - **Reduce 1 FAD to form 1 $FADH_2$.**



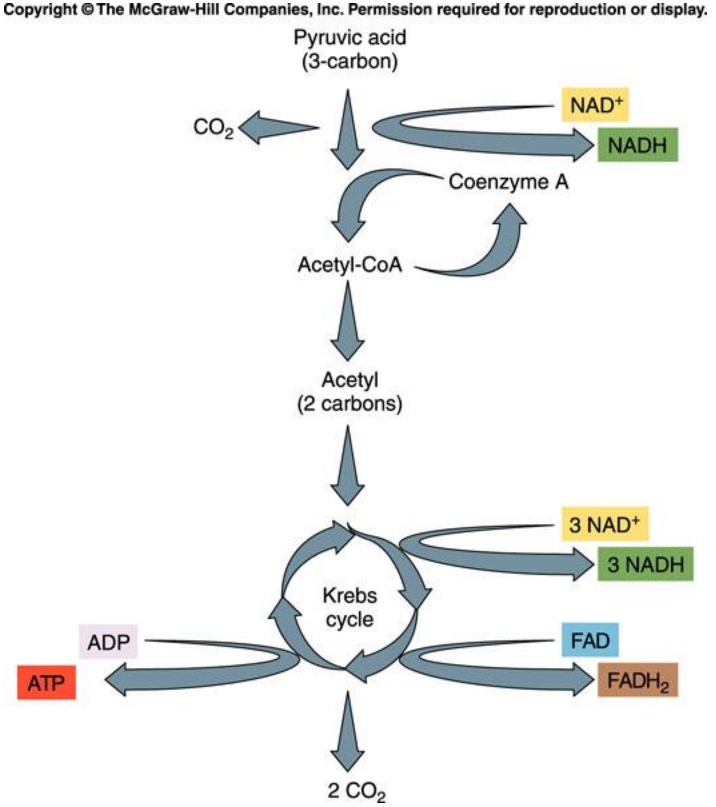
Kreb's Cycle: Summary

Summary of Changes as Pyruvic Acid is Converted to Acetyl-CoA

$$\text{Pyruvic Acid} + NAD^+ + \text{Coenzyme A} \longrightarrow CO + NADH + \text{Acetyl-CoA}$$

Fundamental Summary of One Turn of the Krebs Cycle

$$\underset{(CH_3OC\text{-}CoA)}{\text{Acetyl-Coenzyme A}} + ADP + 3\ NAD + FAD \longrightarrow CoA + ATP + 2\ CO_2 + 3\ NADH + FADH_2$$



Kreb's Cycle: Total yield

- Knowing that two pyruvic acid molecules are converted, the total yield of the Kreb's cycle is:
  - **2 ATP**
  - **8 NADH**
  - **2 $FADH_2$**

Enter the electron transport system

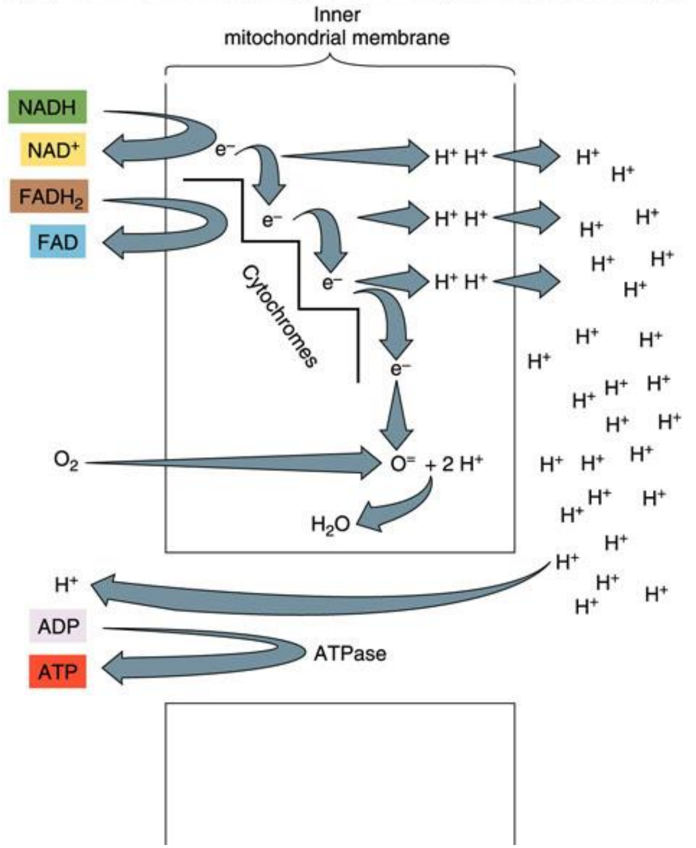
  - **NOTE: Of the 8 NADH produced, 2 are released before acetyl-CoA enters the Kreb's cycle**

Electron-Transport System

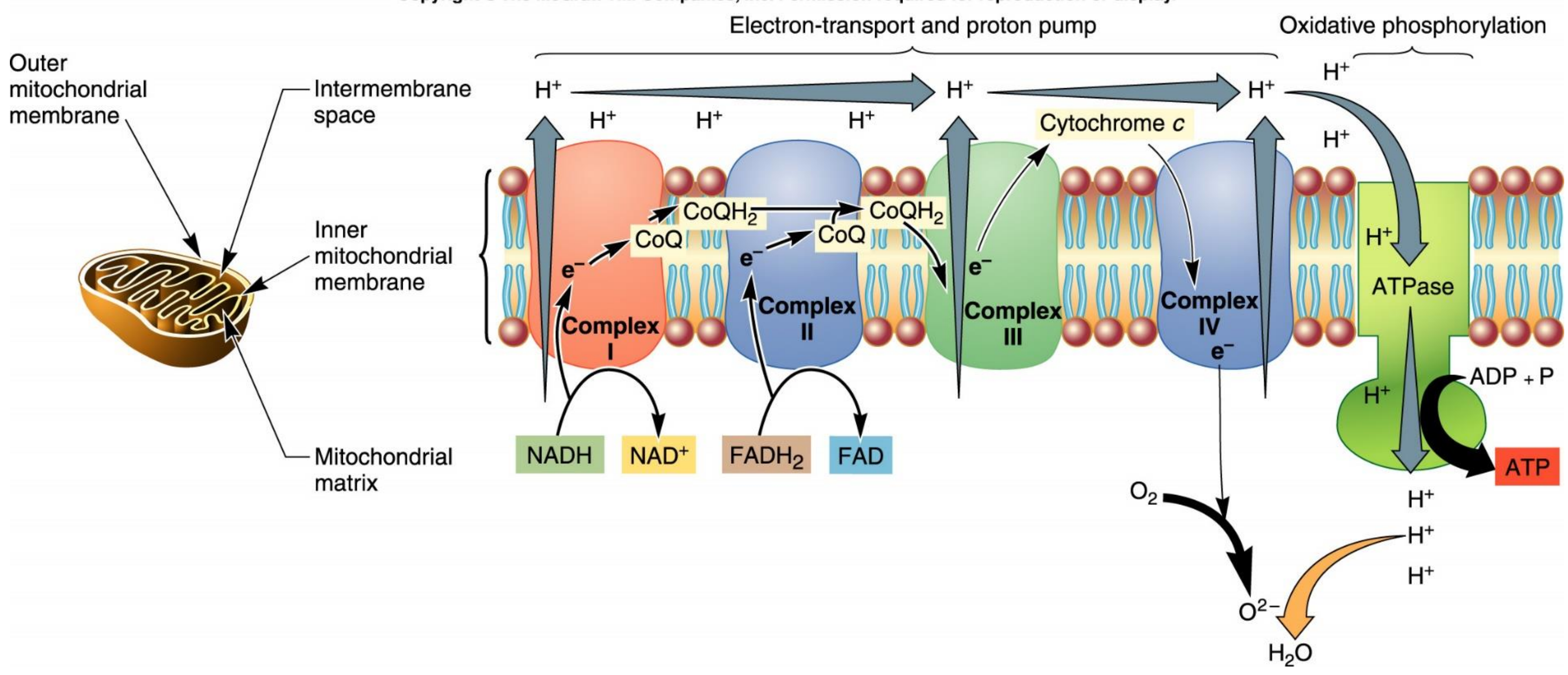
- **NADH and $FADH_2$ release the electrons** they received during glycolysis and the Kreb's cycle to the electron transport chain (ETC).
- The proteins of the ETC transfer the electrons and use the **energy** released to **pump protons**.
  - Protons are pumped from the matrix to the intermembrane space.
  - Creates a **concentration gradient**

Electron-Transport System

- **Oxygen is the final electron acceptor** at the end of the ETC.
  - Oxygen accepts the electrons, combines with protons and becomes **water**.
- The accumulated protons diffuse back into the matrix through **ATP synthase**.
  - The energy released from the diffusion fuels the formation of **ATP**.



The Details of the Electron Transport System (not to memorize)



Total Yields for Aerobic Cellular Respiration per Glucose Molecule

- Glycolysis
  - 2 ATP
  - 2 NADH (converted to 2 $FADH_2$)
- Kreb's cycle
  - 2 ATP
  - 8 NADH
  - 2 $FADH_2$
- Electron transport chain
  - Each NADH fuels the formation of 3 ATP.
    - 8 NADH x 3 ATP = 24 ATP
  - Each $FADH_2$ fuels the formation of 2 ATP.
    - 4 FADH2 x 2 ATP = 8 ATP
- Total ATP=2+2+24+8=36 ATP made from the metabolism of one glucose molecule.

Aerobic Respiration in Prokaryotes

- Very similar to aerobic respiration in eukaryotes
- Since prokaryotes have no mitochondria, it all occurs in the cytoplasm.
- In eukaryotes, the electrons released during glycolysis are carried by NADH and transferred to FAD to form $FADH_2$ in order to get the electrons across the outer membrane of the mitochondrion.
- Because $FADH_2$ results in the production of fewer ATPs than NADH, there is a cost to the eukaryotic cell of getting the electrons into the mitochondrion. This transfer is not necessary in prokaryotes , so they are able to produce a theoretical 38 ATPs for each glucose metabolized, rather than the 36 ATPs produced by eukaryotes

TABLE 6.2

Aerobic ATP Production: Prokaryotic vs. Eukaryotic Cells

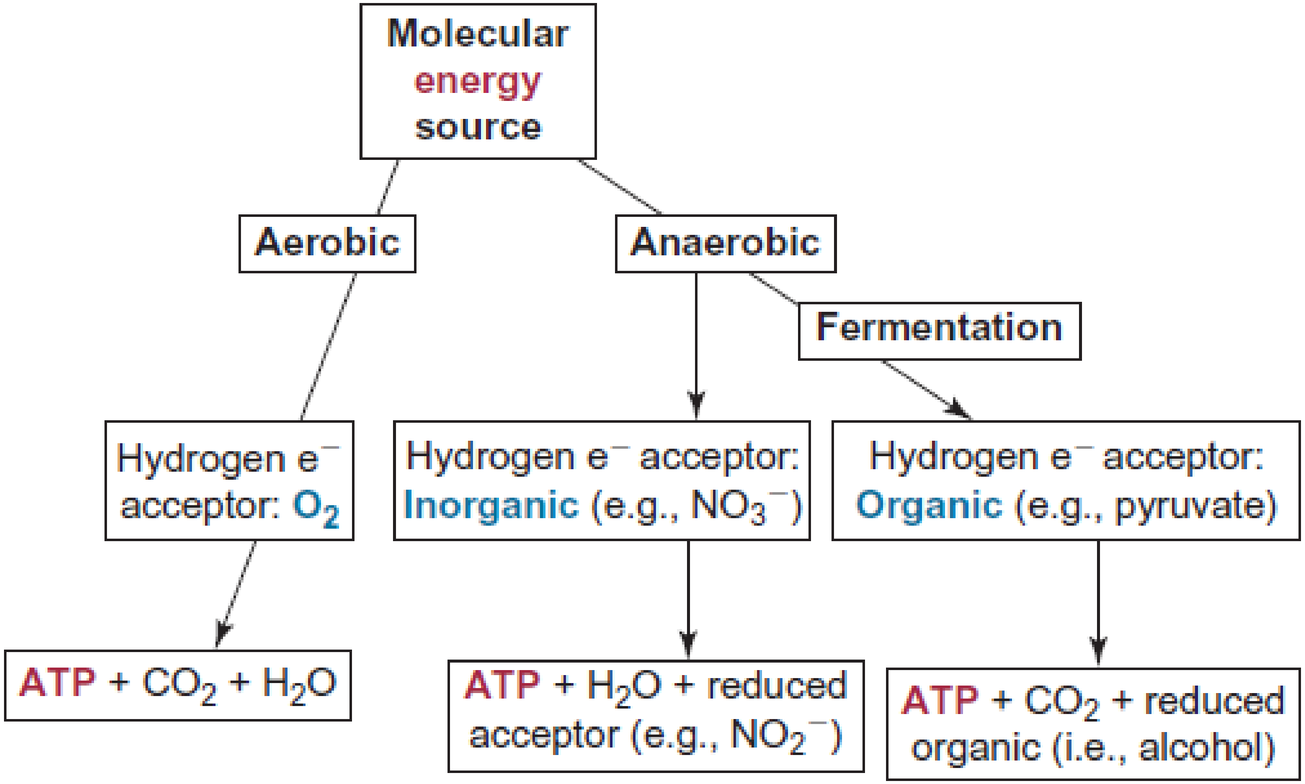
Cellular Respiration Stage	Prokaryotic Cells ATP Theoretically Generated	Eukaryotic Cells ATP Theoretically Generated
Glycolysis	Net gain 2 ATP	Net gain 2 ATP
Krebs cycle	2 ATP	2 ATP
ETS	34 ATP	32 ATP
Total	38 ATP	36 ATP

Anaerobic Cellular Respiration

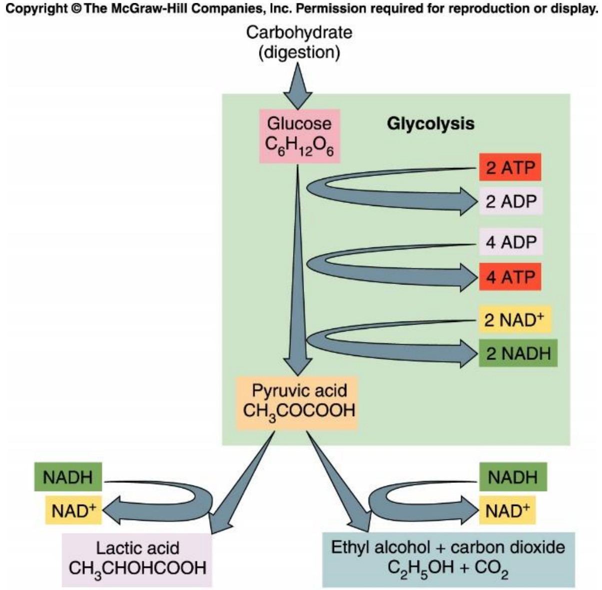
- Some organisms **do not have the enzymes for Kreb's cycle or the electron transport system**
- Some organisms can metabolize glucose in the **absence of oxygen**.
  - Bacteria, Archaea, some fungi, some eukaryotic cells
- An organism that does not require O_2 as its final electron acceptor is called anaerobic and performs **anaerobic cellular respiration**
  - They use inorganic or organic molecules as their final acceptor
- Anaerobic respiration is an **incomplete oxidation** and results in the production of **smaller electron-containing molecules** and energy in the form of **ATP and heat**

Anaerobic Cellular Respiration

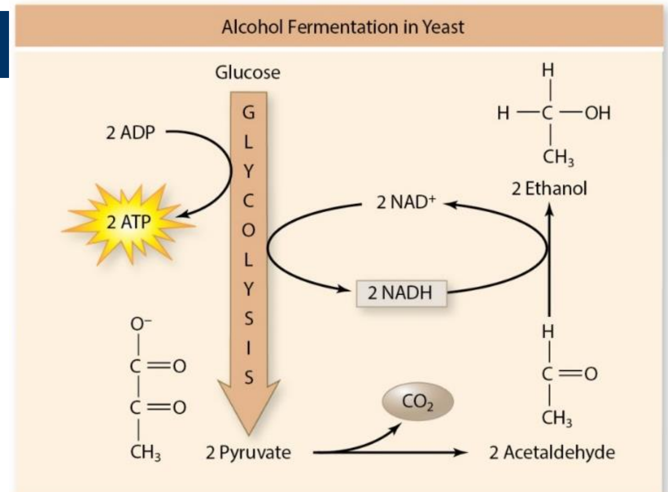
- **Fermentation**
  - anaerobic pathways that **oxidize glucose** to generate ATP by using an **organic molecule as the ultimate electron acceptor**.
  - Electrons removed from sugar in the earlier stages of glycolysis are added to the **pyruvic acid** formed at the end of glycolysis.
  - Depending on the kind of organism and the specific enzymes it possesses, the pyruvic acid can be converted into **lactic acid, ethyl alcohol, acetone, or other organic molecules**
  - The formation of molecules such as alcohol and lactic acid is necessary to **regenerate the NAD+** needed for continued use in glycolysis.

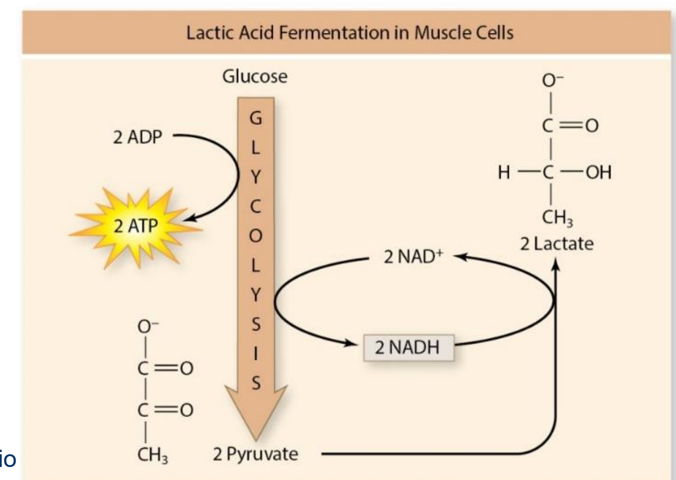

Molecular energy source
Aerobic
Anaerobic
Fermentation
Hydrogen e⁻ acceptor: O2
Hydrogen e⁻ acceptor: Inorganic (e.g., NO3⁻)
Hydrogen e⁻ acceptor: Organic (e.g., pyruvate)
ATP + CO2 + H2O
ATP + H2O + reduced acceptor (e.g., NO2⁻)
ATP + CO2 + reduced organic (i.e., alcohol)

Types of Fermentation



Fermentation Product	Possible Source	Importance
Lactic acid	Bacteria: *Lactobacillus bulgaricus*	Aids in changing milk to yogurt
	Homo sapiens Muscle cells	Produced when O_2 is limited; results in pain and muscle inaction
Ethyl alcohol $+CO_2$	Yeast: *Saccharomyces cerevisiae*	Brewing and baking





Alcoholic Fermentation

- **Starts with glycolysis**
  - Glucose is metabolized to pyruvic acid.
  - A net of 2 ATP is made.
- During alcoholic fermentation
  - Pyruvic acid is reduced to form **ethanol**.
  - **Carbon dioxide** is released.
- **Yeasts** do this
  - Leavened bread
  - Sparkling wine

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Lactic Acid Fermentation

- **Starts with glycolysis**
  - Glucose is metabolized to pyruvic acid.
  - A net of 2 ATP is made.
- During lactic acid fermentation
  - Pyruvic acid is reduced to form **lactic acid**.
  - **No carbon dioxide** is released.

Lactic Acid Fermentation

- **In bacteria**, the lactic acid is a waste product that is used to make yogurt, cultured sour cream, cheeses, and other **fermented dairy products**.
  - The lactic acid makes the milk protein coagulate and become pudding-like or solid. It also gives the products their tart flavor, texture, and aroma
- **Muscle cells** have the enzymes to do this, but brain cells do not.
  - Muscle cells can survive brief periods of oxygen deprivation, but brain cells cannot.
  - Lactic acid "burns" in muscles and makes muscles tired when we exercise

Metabolizing Other Molecules

- Cells will use the energy in carbohydrates first.
  - Complex carbohydrates are metabolized into simple sugars.
- Cells can use the energy in fats and proteins as well.
  - Fats are digested into fatty acids and glycerol.
  - Proteins are digested into amino acids.

Metabolizing Other Molecules

- The basic pathways organisms use to extract energy from fat and protein are the same as for carbohydrates: **glycolysis, the Krebs cycle, and the electron-transport system.**

- However, there are some **additional steps** necessary to get fats and proteins ready to enter these pathways at several points in glycolysis and the Krebs cycle where fats and proteins enter to be respired.

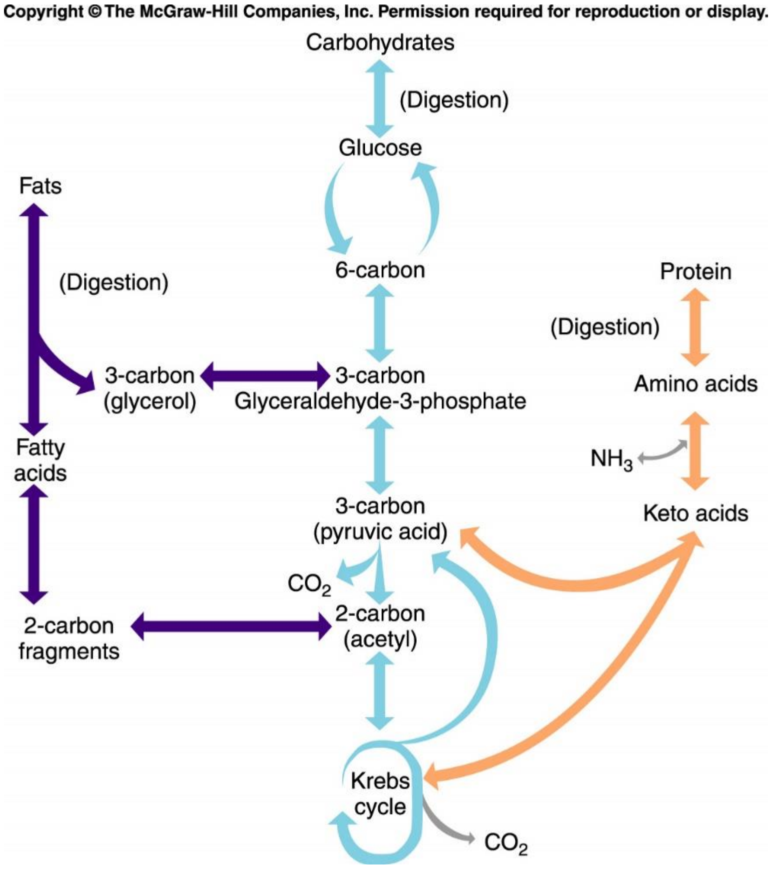
Fat Respiration

- Fats are broken down into
  - Glycerol
  - Fatty acids
- Glycerol
  - Converted to glyceraldehyde-3-phosphate
  - Enters glycolysis
- Fatty acids
  - Converted to acetylCoA
  - Enter the Kreb's cycle
- Each molecule of fat fuels the formation of many more ATP than glucose.
  - This makes it a good energy storage molecule.

Protein Respiration

- Proteins are digested into amino acids.
- Then amino acids have the amino group removed.
  - Generates a keto acid (acetic acid, pyruvic acid, etc.)
  - Enter the Kreb's cycle at the appropriate place

The Interconversion of Fats, Carbohydrates and Proteins



The Bottom Line

- Carbohydrates, fats and proteins can all be used for energy.
  - Glycolysis and the Kreb's cycle allow these types of molecules to be interchanged.
- If more calories are consumed than used
  - The excess food will be stored.
  - Once the organism has all of the proteins it needs
    - And its carbohydrate stores are full
    - The remainder will be converted to and stored as fat.